

ITUMELENG NTELA

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PROFESSIONAL SUMMARY

BSc Information Technology degree holder pursuing Honours in Computer Science and Information Technology at North-West University. Skilled in full-stack development, cloud computing, artificial intelligence, data engineering, and cyber security. Proven track record of building and developing solutions using reinforcement learning algorithms to cloud-based threat intelligence platforms. Strong background in academic leadership and self-development. Looking for opportunities in software engineering, data analytics, cloud computing, AI, and IT risk management.

EDUCATION

BSc Honours in Computer Science and Information Technology | (In Progress)

North-West University

BSc in Information Technology | 2023 - 2025 (Completed)

North-West University

- Golden Key International Honour Society - Top 15%

TECHNICAL SKILLS

Languages	Python, Java, C#, C++, JavaScript, SQL
AI & ML	Reinforcement Learning (Q-Learning), Machine Learning, Microsoft AI Fluency
Cloud & Azure	Google cloud, Microsoft Azure, AWS Fundamentals
Data & BI	Apache Kafka, Power BI, Data Analytics, Data Structures, ETL Pipelines
Web / Full-Stack	HTML, CSS, JavaScript, Node.js, React, REST APIs
Databases	SQL Server, MongoDB, Azure Cosmos DB, NoSQL, Data Modelling
Security	Cybersecurity Fundamentals, Threat Intelligence, IT General Controls, Network Security
Frameworks	.NET Framework, WinForms, Visual Studio, VS Code, PyCharm, Git
Methodologies	Agile, Scrum, System Analysis & Design, CI or CD Principles, Requirements Analysis

PROJECTS

Cloud-Based Oncology Management System | 2026 Role: Owner

- Designing and developing a cloud-hosted system to digitise patient records in an oncology setting, reducing reliance on paper-based processes.
- Applying Google cloud services for cloud storage, scalability, and security compliance.
- Conducting requirements analysis to identify workflow inefficiencies and translate them into system specifications.
- **Technologies:** Google cloud, System Analysis & Design, Database Design.

Smart City Unified Threat Intelligence Platform | 2026 Role: Owner

- Designed and built a real-time multi-stream threat detection system processing Azure Event Hub stream.
- Implemented Azure Stream Analytics with event-time windowing.
- Stored enriched event data in Azure Cosmos DB and built Power BI dashboards visualising KPIs.
- **Technologies:** Microsoft Azure, Power BI, Python, SQL, IoT, Real-Time Data Engineering.

Smart Mancala Player - Reinforcement Learning AI Agent | 2026 Role: Owner

- Built a Q-Learning AI agent for the traditional Southern African game Mancala, trained through 50,000 episodes of self-play with epsilon-greedy exploration and exponential decay.
- Engineered a shaped reward function to accelerate convergence and avoid sparse reward problems
- Developed a fully playable web interface supporting Human vs AI, AI vs AI, and hint modes bringing the trained agent to life in a browser.
- **Technologies:** Python, Q-Learning, Reinforcement Learning.

ClearVue BI Platform | 2025 Role: Team Member

- Built a real-time Business Intelligence platform processing live data streams with Apache Kafka.
- Implemented MongoDB for scalable NoSQL data storage and designed Power BI dashboards for executive-level decision making.
- **Technologies:** Apache Kafka, MongoDB, Power BI, Node.js.

StudentDev Hub Platform | 2025 Role: Group Leader

- Led an Agile team to deliver a full-stack web application using Node.js, React and MongoDB, managing sprints, code reviews, and final delivery.
- **Technologies:** Node.js, React, MongoDB, REST APIs, Agile.

Vitality Clinic System | 2024 Role: Group Leader

- Directed development of a Windows desktop application for patient records and appointment management, with structured relational database design.
- **Technologies:** C#, .NET Framework, WinForms, SQL Server.

WORK EXPERIENCE

Peer-Assisted Learning Facilitator - Java (CMPG211) | North-West University · 2025

- Facilitated structured weekly sessions on object-oriented programming and Java fundamentals for groups of students.
- Developed session plans and adapted delivery based on student feedback, demonstrating communication and leadership skills.

Student Learning Assistant - Python (CMPG111) | North-West University · 2025

- Supported students with Python fundamentals, debugging techniques, and computational thinking.
- Delivered one-on-one and group mentoring sessions to improve academic performance.

Student Learning Assistant - C++ (CMPG121) | North-West University · 2024

- Assisted students with C++ syntax, memory management, and problem-solving strategies.
- Conducted individual consultations and group workshops to reinforce lecture content.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

2026 Vodacom Digital Skills Programme - Short Course

2025 Microsoft AI Fluency Certificate

2025 FNB App Academy - Full Stack Development

2025 Udemy - Data Analysis Bootcamp

2024 Chongqing University of Technology - Cyber Security Training

References available on request